

Secondary-Power-Cell-Aware Detailed Placement in Multiple Power Domain Designs

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Abstract—Using multiple power domains is a common technique to achieve significant power savings in modern designs. However, providing power to *cross-domain cells* is challenging at advanced technology nodes, due to strict IR drop constraints and competition for routing resources between secondary power routing and signal routing. Despite its importance, this issue has received limited attention in recent literature. In this paper, we propose a detailed placement method that accounts for secondary power routing. Our approach employs an integer linear programming (ILP) model, complemented by targeted adjustments to the placement of cross-domain cells, to optimize the efficiency of secondary power routing. We validate our results with commercial design tools, showing that our approach effectively improves secondary power routing performance by reducing the wirelength of the secondary power distribution network (PDN) and decreasing violations due to secondary PDN constraints.

I. INTRODUCTION

Low power has become one of the most important features in many electronic products, with multiple supply voltages (MSVs) and power domains being widely used to achieve reduced power consumption. MSVs control circuit sections independently to reduce dynamic and leakage power, while power gating boosts efficiency by shutting off idle domains. In multiple power domain designs, cross-domain cells, such as always-on buffers/inverters, retention flip-flops, and isolation cells, are essential for enabling power domain shutdowns to achieve power savings. These cells serve crucial roles, such as level shifting, transmitting default values, and ensuring that the design continues to function correctly even when certain power domains are turned off. Through these functions, cross-domain cells support the seamless operation of multiple power domain designs under varying power conditions.

These cells have both a *primary power pin*, shown in light blue of left cell in Fig. 1, and a *secondary power pin*, shown in gray. The primary power pin connects to the *primary power delivery network* (primary PDN), supplying power from the primary power source during regular operation, shown in pink net. Meanwhile, the secondary power pin connects to the *secondary power delivery network* (secondary PDN) to provide power when the primary power domain is shut down, enabling continuous operation of the cross-domain cells, shown in dark blue nets. Each cross-domain cell belongs to one power domain but also connects to a secondary PDN, ensuring the independence between the two power domains, both of which supply power to the cross-domain cell as needed. In this paper, we refer to the ordinary power-supplying power domain of a cross-domain cell as the *primary power domain* (PPD), while the secondary PDN and all cells connected to it are collectively referred to as the *secondary power domain* (SPD). The term *power domain* will refer broadly to the combined set of both primary and secondary power domain.

In multiple power domain designs, when cells from different power domains or cross-domain cells are present within the same block, it

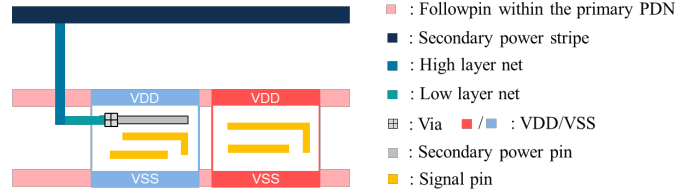


Fig. 1. The architecture of the secondary PDN and cross-domain cells. The blue cell represents a cross-domain cell connected to the secondary PDN, while the red cell represents a general cell.

is necessary to interleave power stripes for each power domain [11]. This arrangement allows each cell to quickly connect to the appropriate power stripe for its domain. Additionally, a design can contain multiple SPDs, with each SPD having its own independent secondary PDN that does not connect to others. Each cross-domain cell connects to only one secondary PDN, thus belonging to a single SPD as a backup power source. As shown in Fig. 2(a), the orange section represents the primary power stripe for the area, while the blue and green stripes represent power stripes for two different SPDs. This interleaved configuration helps cells connect more efficiently to their designated power stripes, reducing wire length and connection complexity. In addition, Fig. 2(b) shows the conventional way of connecting the cells to the primary PDN. Since all cells in this area share the same primary power domain, regular power and ground rails are used. Figs. 2(c) and (d) respectively illustrate the secondary PDN and cross-domain cells belonging to SPD 1 and 2, respectively. The secondary PDNs connect the gray secondary power pins of cross-domain cells to the secondary power stripes. Note that the trunk-based secondary PDN structures are preferred, which enhance IR drop performance and power stability and decrease routing complexity and the likelihood of routing violations. Each orange pin represents a signal pin, which is not connected during the secondary power routing phase but may suffer from violations if covered by some secondary PDN.

Multiple power domain designs have been widely explored in the literature. Studies such as [9], [10], [11] have researched PDN optimization for multiple power domain designs. More recent work has also examined various power-aware considerations in placement and floorplanning domains [12], [13]. However, these studies primarily focus on primary PDN and do not extend their considerations to secondary PDN. In contrast, research addressing cross-domain cells has largely focused on power gating transitions [7], [8], buffering within cross-domain designs [6], [5], and rule-checking [4]. However, they lack emphasis on placement or routing for PDN optimization. To the best of our knowledge, the only existing study specifically addressing secondary power mesh improvements is [14]. This work develops an IR drop prediction model and a design rule violation prediction model to guide the construction of sub-meshes for cross-domain cells. However, the approach only considers mesh-based secondary PDNs, consuming more routing resources for the low-density regions. In addition, [14] optimizes secondary PDNs for a given cell placement result; however, the placement of cross-domain cells could greatly affect both the PDN complexity and power performance.

This work was partially supported by TSMC, Synopsys, MPI, and NSTC of Taiwan under Grant No's NSTC 113-2927-I-011-502, 113-2640-E-002-001, 113-2640-E-006-001, 113-2222-E-011-005-MY3, and 114-2927-I-011-501.

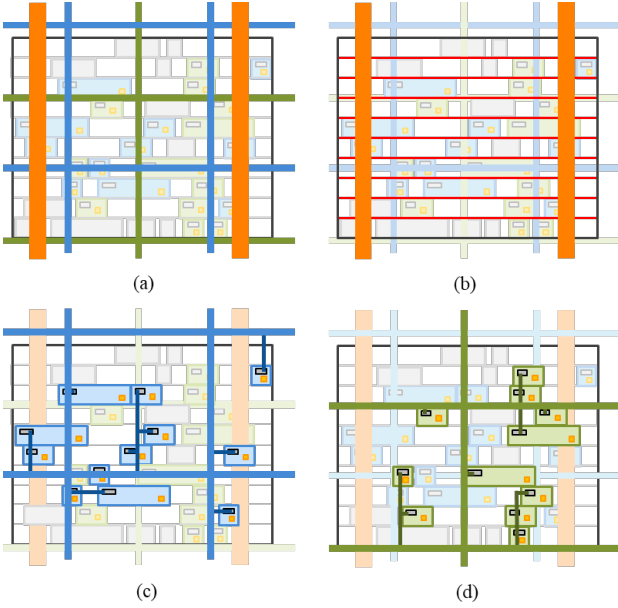


Fig. 2. An example of cross-domain cells and secondary PDN architecture. (a) The structure of power stripes in the design, where the orange stripes represent the primary PDN, and the blue and green stripes represent two different SPDs. (b) Primary PDN within this power domain. (c) The PDN and cells in SPD 1. (d) The PDN and cells in SPD 2.

In this paper, we propose a secondary power routing-aware detailed placement to refine the positions of cross-domain cells, facilitating better secondary power routing with a commercial tool. The main contributions of this paper are as follows:

- This is the first study to address reducing violations caused by secondary PDNs and minimizing wirelengths of tree-based secondary PDNs in the placement stage.
- We propose an integer linear programming (ILP) secondary-power-routing-aware placement refinement algorithm, followed by a trapped cell relocation technique and a pin alignment procedure.
- The proposed placement algorithm is validated by using a commercial tool, and the results demonstrate that the optimized placements reduce the critical secondary PDN wirelength by 11%, the total secondary PDN wirelength by 30%, and the number of violations by over 66%.

The remainder of this paper is structured as follows: Section II introduces the key challenges of secondary power routing and defines the problem. Section III details the proposed solution and algorithm. Section IV presents the reductions in wirelength and design rule violations achieved through our placement approach, and Section V concludes the study.

II. POTENTIAL ISSUES AND SOLUTION CONCEPTS FOR SECONDARY POWER ROUTING

In the following, Section II-A will discuss the negative effects that secondary PDNs introduce to the design, along with the physical factors that contribute to these issues. In Section II-B, we will present conceptual solutions and strategies to address these problems, and clearly define the problem at the end of the section.

A. Causes of Violations due to Secondary PDNs

Fig. 3(a) provides an example showing how design rule violations can be caused due to the connection patterns of secondary PDNs. In the figure, the blue and green sections represent two independent SPDs, each with its secondary PDN and cross-domain cells. The gray part within each cell indicates the secondary power pin, while the orange part represents the signal pin. Because the typical routing order begins with primary PDN, followed by secondary power routing and then

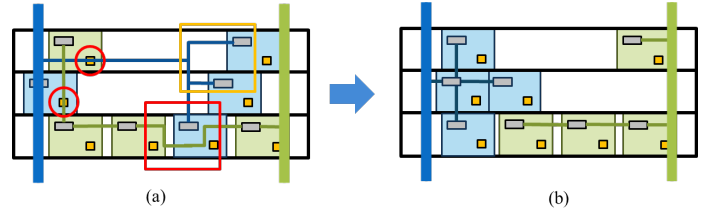


Fig. 3. Examples of potential non-ideal scenarios involving multiple secondary PDNs. (a) A given placement instance. (b) The optimized placement after cell rearrangement.

signal routing, the routing of secondary PDNs can directly impact signal routing, resulting in violations without the awareness of signal nets.

The resulting violations fall into two main types: the first occurs when connections in secondary power routing cannot be completed, and the second arises when secondary PDNs interfere with signal routing. Connections for secondary PDNs typically do not present issues due to the availability of complete routing resources before signal routing. In contrast, signal routing can greatly be affected by secondary PDNs due to its lower priority. The primary reasons for signal routing failure due to secondary PDN are as follows. First, if a secondary PDN directly covers a signal pin, it prevents any connection to the signal pin, resulting in a violation, as shown by the two red circles in Fig. 3(a). The second reason is that if the secondary PDN occupies too many routing resources, the signal network may lack sufficient resources for routing. This situation can lead to an increase in signal network wire length and even cause incomplete routing with violations.

To address this problem, rearranging cross-domain cells associated with different secondary PDNs can serve as a practical solution. If cells belonging to the same secondary domain are spaced apart, detours can be caused during secondary power routing. For instance, as illustrated in Fig. 3(a), the green cells within the red box are separated by the blue cell, leading to a detour in the green secondary PDN. In addition, if a cross-domain cell is surrounded by cells from different secondary PDNs, it could be obstructed from connecting to its own secondary power stripes, resulting in violations. Therefore, by grouping cells belonging to the same secondary domain together and placing each cell close to its *target secondary power stripe (TSPS)*, detours and obstructions could be effectively reduced, thereby improving both the difficulty and quality of secondary power routing. The TSPS of each cell is simply the closest secondary power stripe to its secondary power pin. Moreover, the arrangement of secondary power pins also plays a crucial role. It is generally desirable to minimize the number of wire bends in PDN routing to facilitate easier connections and power performance. For instance, as the blue cells within the orange box in Fig. 3(a), aligning their secondary power pins as shown in Fig. 3(b) can significantly reduce the secondary PDN wirelength and detours, thereby enhancing the likelihood of successful routing.

In addition to violations, secondary PDN wirelength significantly affects design performance, and both the *critical secondary PDN wirelength* (WL_{sec}^c) and the *total secondary PDN wirelength* (WL_{sec}^t) should be minimized. WL_{sec}^c refers to the wirelength of the longest secondary PDN path, impacting IR drop and performance, while WL_{sec}^t is the total wirelength of all secondary PDNs. Excessive WL_{sec}^t can degrade power performance and consume resources needed for subsequent signal routing, leading to longer signal wirelengths, increased routing difficulty, and more violations. Based on these factors, the objective function for improving secondary PDNs can be expressed as follows:

$$\text{Minimize} \quad \#vios + \alpha \cdot WL_{sec}^t + \beta \cdot WL_{sec}^c + \gamma \cdot WL_{sig}, \quad (1)$$

where $\#vios$ represents the number of violations, and WL_{sig} represents the total wirelength of signal nets. α , β , and γ are user-defined

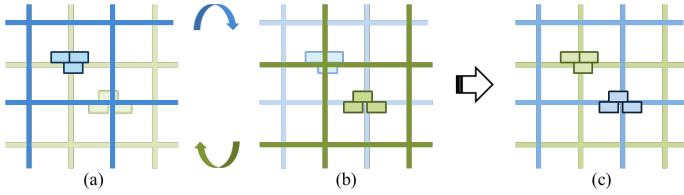


Fig. 4. Diagram of critical cell placement toward its TSPS. (a) Critical cells and power stripe of SPD 1. (b) Critical cells and power stripes of SPD 2. (c) Ideal placement after swapping the locations of the critical cells in (a) and (b).

constants, typically much less than one ($\alpha, \beta, \gamma \ll 1$), to prioritize violation reduction.

B. Operations for Placement Optimization

Equation (1) represents the objective function for the entire design, but it cannot be directly applied to the placement algorithm, as the number of violations and wirelengths cannot be directly derived during placement. Therefore, we explain how to achieve approach this goal using a placement algorithm. As mentioned above, the primary way to eliminate violations is to move cross-domain cells closer to their TSPSs. This approach has two main benefits: it reduces secondary PDN wirelength and aggregates the cells of the same secondary power domain closer together. In addition, after examining multiple designs, we found that connections that can greatest impact signal routing are usually the cross-domain cells farthest from their TSPSs. We define such cells as *critical cross-domain cells* (referred to as *critical cells* hereafter). Critical cells are identified as the top- x cells with the longest distances to their TSPSs.

Three operations can be used to rearrange the cross-domain cells for secondary PDNs. First, it is straightforward to directly move a critical cell c_i with large ζ_i toward its TSPS. Besides, since each cross-domain cell has a different TSPS, using cell swapping for placement has the potential to yield a win-win result. Fig. 4 shows an example, where Figs. 4(a) and (b) represent two independent SPDs, and their critical cells lie near the power stripes of the other secondary power domain. Directly swapping the cells from different secondary power domains results in a much better placement result, allowing both to be positioned closer to their respective TSPSs, as illustrated in Fig. 4(c). Furthermore, in addition to reducing the distance between a cell and its TSPS, secondary power routing difficulty can also be minimized by aligning secondary power pins or relocating cells that are surrounded by cells from other SPD. The detailed explanations of adopting these operations in our methodologies will be provided in Section III.

C. Problem Formulation

Given a multiple power domain design consisting of primary PDN, a set of secondary PDNs, a set of standard cells, a set of cross-domain cells, a legalized placement result, and a netlist containing all connection relationships, the objective of this work is to perform secondary power routing-aware placement refinement for the design to minimize the total secondary PDN wirelength, the critical secondary PDN wirelength, the total number of design rule violations, and the overhead in signal wirelength.

III. PROPOSED FLOW

In the following section, we will provide a detailed description of how we use placement strategies to facilitate secondary power routing and signal routing and explain how we validate the effectiveness of the entire approach.

A. Secondary Power Optimization Framework

Fig. 5 illustrates the flowchart of our algorithm, divided into two main parts: the secondary-power-routing-aware placement stage (placement stage) and the routing and verification stage (routing stage). The

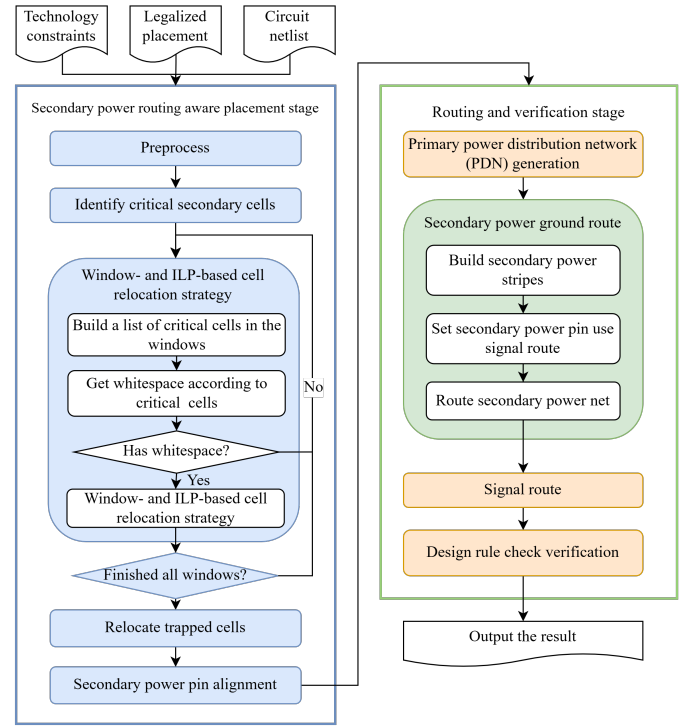


Fig. 5. The overall flow of the proposed framework

placement stage focuses on detailed placement to address violations caused by secondary PDNs, while the routing stage uses a commercial tool for secondary power routing and design rule check (DRC) to validate the effectiveness and feasibility of the placement algorithm.

The flow begins with data loading and preprocessing to establish data structures, followed by identifying critical cells among cross-domain cells based on their distances to their TSPSs. These cells undergo placement with a window-based secondary-power-routing-aware placement refinement procedure with an ILP formulation. Next, all cross-domain cells are processed to relocate trapped cells, and secondary power pins are aligned to ensure low-complexity routing patterns. In the routing stage, we use a commercial tool to generate the primary PDN, followed by secondary PDN generation. Signal routing is then performed to evaluate the impact of cross-domain cell replacement and secondary power routing. The final step is DRC verification, assessing the effectiveness of the proposed methodologies in resolving secondary power routing issues and providing quantitative comparison data. We detail each major invention of this work on placement refinement in the following subsections.

B. Window- and ILP-based Cell Relocation Strategy

In this stage, our primary goal is to place critical cells near their respective TSPSs. The ILP-based algorithm operates in an iterative manner, identifying critical cells and available white spaces within each window and pairing them for re-placement in a fill-in-the-blank style until all iterations are complete. Each iteration has a set time limit for solving to avoid runtime explosion. Details of each component are explained below.

1) *Critical Cell and White Space Identification*: The definition of critical cells is based on the distance between cross-domain cells and their TSPSs. In this study, we identify the top 25% of cells with the longest distances as critical cells for subsequent placement refinement. We define a *white space* as an empty row segment, which includes unoccupied areas and the original positions of critical cells. One of the advantages of considering the original positions of critical cells is to enable multi-way cell swapping. In addition, this characteristic allows the ILP-based algorithm to ensure that the solution space encompasses the original design, guaranteeing that the resulting performance is not worse than that of the initial design.

TABLE I
SYMBOLS USED IN THE WINDOW- AND ILP-BASED CELL RELOCATION
STRATEGY

Variable	Definition
C^{cri}	The set of critical cells.
c_i	The i -th cell in C^{cri} .
N	The set of nets.
N_i	The set of nets connected to c_i .
n_{ij}	The j -th net in N_i .
$HPWL_{inc}$	Total increased HPWL of signal nets connected to critical cells.
Z	Total distance of all critical cells to their TSPS.
α	Ratio number of $HPWL_{inc}$ and Z .
x_i/y_i	The coordinate of a location candidate of c_i .
M_{xij}/M_{yij}	The maximum x/y coordinate values among cells connected to n_{ij} .
m_{xij}/m_{yij}	The minimum x/y coordinate values among cells connected to n_{ij} .
$incX_i/incY_i$	The increase in HPWL in the x/y direction caused by c_i .
$d_i^t/d_i^b/d_i^l/d_i^r$	Distance from secondary power pin of c_i to the top/bottom/left/right secondary power stripe.
$b_i^t/b_i^b/b_i^l/b_i^r$	A Boolean variable used to determine whether c_i connects to the top/bottom/left/right secondary power stripe.
ζ_i	The distance between the critical cell i and the TSPS.
$p_i^t/p_i^b/p_i^l/p_i^r$	The y- or x-coordinate of the top/bottom/left/right secondary power stripe of c_i .
x_i^{sec}/y_i^{sec}	The location of cell c_i 's secondary power pin.
M	A large constant.
S	The set of white spaces.
s_k	The k -th white space in S .
$e_{i,k}$	A Boolean variable to determine whether c_i is in s_k .
row_i^c	The row of c_i .
row_k^s	The row of s_k .
$\min X_k^s$	Minimum x-values of s_k .
$\max X_k^s$	Maximum x-values of s_k .
w_i^c	The width of c_i .
w_k^s	The width of s_k .
$d_{i,k}$	A Boolean variable. $d_{i,k} = 0$ indicates row_i^c and row_k^s is equal; $d_{i,k} = 1$, otherwise.
$g_{i,j}$	A Boolean variable. $g_{i,j} = 0$ indicates c_i is on the right side of c_j ; $g_{i,j} = 1$, otherwise.

2) *ILP Objective Function:* After identifying the critical cell list and the white space list, we can proceed to match them using the ILP algorithm. Table I presents the notations and descriptions used in the ILP formulation, and Equation (2) shows the objective function:

$$\text{Minimize} \quad \alpha \times HPWL_{inc} + (1 - \alpha) \times Z. \quad (2)$$

In the placement process, we utilize HPWL (Half-Perimeter Wire Length) to model the wirelength of signal nets, and we denote the total increase in signal wirelength due to placement refinement as $HPWL_{inc}$. Z is calculated as the total straight-line distance from the center of the secondary power pin of each critical cell to the center of its TSPS, representing the secondary PDN wirelength in this stage. The $HPWL_{inc}$ can be expressed by the following:

$$HPWL_{inc} = \sum_{c_i \in C^{cri}} (incX_i + incY_i), \quad (3)$$

$$incX_i = \sum_{n_{ij} \in N_i} \left(\frac{|x_i - M_{xij}| + |x_i - m_{xij}| - M_{xij} + m_{xij}}{2} \right), \quad (4)$$

$$incY_i = \sum_{n_{ij} \in N_i} \left(\frac{|y_i - M_{yij}| + |y_i - m_{yij}| - M_{yij} + m_{yij}}{2} \right). \quad (5)$$

Equations (4) and (5) respectively compute the overall HPWL increase in the x- and y-direction of all the nets connected to a cell c_i when moving c_i from its original position to a location candidate at (x_i, y_i) . Note that no matter whether c_i is moved to a location inside or outside

the original net bounding box, the increase in HPWL can be accurately computed using the equations. Next, the total straight-line distance of all secondary power pins to their TSPSs Z can be computed by the following equation:

$$Z = \sum_{c_i \in C^{cri}} \zeta_i, \quad (6)$$

and ζ_i can be expressed as follows:

$$\zeta_i = b_i^t \times d_i^t + b_i^b \times d_i^b + b_i^l \times d_i^l + b_i^r \times d_i^r, \quad (7)$$

$$b_i^t + b_i^b + b_i^l + b_i^r = 1. \quad (8)$$

Equation (7) is used to compute distance from c_i to its TSPS. Only one of b_i^t , b_i^b , b_i^l , and b_i^r will be 1, indicating that this power stripe is the TSPS for c_i , while the others will be 0. Therefore, we constrain this with Equation (8). It is important to note that, since this objective function is meant for use by an ILP solver, variable multiplications are not allowed. Therefore, we need to reformulate the above mathematical expression in an equivalent manner.

The primary task in the following is to express $d \cdot b = \zeta$ in a way that does not involve the multiplications of variables, and then to apply this mathematical expression to replace Equation (7). We try to express $d \cdot b = \zeta$ as inequalities, where $d, \zeta \in \mathbb{Z}$ and b is a binary variable, ζ represents the distance between the cell and the power stripe. This implies that when $b = 0$, ζ must equal 0, and when $b = 1$, ζ must equal d . We can utilize the big-M method to resolve this issue, and the transformed inequality is as follows:

$$\zeta \leq d, \quad (9)$$

$$\zeta \leq M \times b, \quad (10)$$

$$\zeta \geq 0, \quad (11)$$

$$\zeta \geq d - M(1 - b), \quad (12)$$

where M is a large constant much greater than ζ and d , and $M \in \mathbb{Z}$. The above equations guarantee that when $b = 0$, ζ can be constrained to 0 by Equation (10) and Equation (11). Conversely, when $b = 1$, ζ can be constrained to d by Equation (9) and Equation (12). Given that our objective function seeks to minimize the value, it is unnecessary to impose upper bounds from Equation (9) and Equation (10); it suffices to restrict the lower bounds using Equation (11) and Equation (12) only. Based on the above derivations, we can incorporate this concept into Equation (7) to obtain the following inequalities that satisfy ILP requirements:

$$\zeta_i \geq |p_i^t - y_i^{sec}| - M(1 - b_i^t), \quad (13)$$

$$\zeta_i \geq |y_i^{sec} - p_i^b| - M(1 - b_i^b), \quad (14)$$

$$\zeta_i \geq |x_i^{sec} - p_i^l| - M(1 - b_i^l), \quad (15)$$

$$\zeta_i \geq |p_i^r - x_i^{sec}| - M(1 - b_i^r), \quad (16)$$

$$\zeta_i \geq 0. \quad (17)$$

The distance d between the cell and the power stripe will vary based on their relative position. Specifically, if the power stripe is located above, below, left, or right of the cell, the distance can be calculated as $|p_i^t - y_i^{sec}|$, $|y_i^{sec} - p_i^b|$, $|x_i^{sec} - p_i^l|$, or $|p_i^r - x_i^{sec}|$, respectively. Only one of the right-hand sides of Equations (13)–(16) will be greater than zero, while the others will be greater than a negative value, making them negligible. Equation (17) can be omitted by defining ζ_i to be no less than 0 in the implementation.

3) *ILP Constrain Function:* In the following, we consider the fundamental row-based design rule constraints to ensure that cells are placed reasonably within the available white spaces. Specifically, cells must not overlap, must be placed within rows, and must be aligned with sites. The first set of constraints consists of five equations, primarily used for matching critical cells to available white spaces:

$$\sum_{s_k \in W} e_{i,k} = 1, \quad \forall c_i \in C^{cri}, \quad (18)$$

$$\sum_{\forall s_k \in S} (e_{i,k} \times \text{row}_k^s) = \text{row}_i^c, \quad \forall c_i \in C^{\text{cri}}, \quad (19)$$

$$\sum_{\forall s_k \in S} (e_{i,k} \times \min X_k^s) \leq x_i, \quad \forall c_i \in C^{\text{cri}}, \quad (20)$$

$$\sum_{\forall s_k \in S} (e_{i,k} \times \max X_k^s) \geq x_i + w_i^c, \quad \forall c_i \in C^{\text{cri}}, \quad (21)$$

$$\sum_{\forall c_i \in C^{\text{cri}}} (e_{i,k} \times w_i^c) \leq w_k^s, \quad \forall s_k \in S. \quad (22)$$

If $e_{i,k}$ equals 1, it indicates that c_i is placed in s_k ; otherwise, it indicates that c_i is placed in another white space. Equation (18) states that a cell must and can only be placed in one white space, and thus the sum of variables $e_{i,k}$ equals 1. Equation (19) ensures that c_i is placed in the same row as its target white space. In this equation, row_i^c is a variable, while row_k^s is a constant, ensuring that row_i^c is set to row_k^s . Equation (20) and Equation (21) are used to place c_i within the horizontal span of s_k . $\min X_j^w$, $\max X_j^w$, w_i^c , and w_k^s are given constants. The constraints above ensure that cells are accurately matched to their white spaces, placed in the same row, and positioned within the white space. However, if a white space is large enough, it may be able to accommodate more than one cell. Therefore, we use Equation (22) to limit the total width of cells that can be placed within a white space, ensuring that it does not exceed the width of the white space itself, preventing each white space from overflowing. This ensures that all cells paired with s_k can fit within it without exceeding its capacity.

The second set of constraints primarily aims to prevent overlaps between cells, as formulated below:

$$|\text{row}_i^c - \text{row}_j^c| \leq M \cdot d_{i,j}, \quad \forall c_i, c_j \in C^{\text{cri}}, \quad (23)$$

$$|\text{row}_i^c - \text{row}_j^c| \geq d_{i,j}, \quad \forall c_i, c_j \in C^{\text{cri}}, \quad (24)$$

$$x_i + w_i^c \leq x_j + M \cdot (1 - g_{i,j}) + M \cdot d_{i,j}, \quad \forall c_i, c_j \in C^{\text{cri}}, \quad (25)$$

$$x_j + w_j^c \leq x_i + M \cdot g_{i,j} + M \cdot d_{i,j}, \quad \forall c_i, c_j \in C^{\text{cri}}. \quad (26)$$

Equation (23) and (24) are primarily used to determine whether the assigned rows of c_i and c_j are the same. If they are the same, the boolean variable $d_{i,j}$ will equal 0; otherwise, it will equal 1. In Equation (23), when $\text{row}_i^c \neq \text{row}_j^c$, $|\text{row}_i^c - \text{row}_j^c|$ will be greater than 0, meaning that $d_{i,j}$ must be 1 for the inequality to hold. Conversely, when $\text{row}_i^c = \text{row}_j^c$, $d_{i,j}$ can be either 0 or 1, hence Equation (24) ensures that $d_{i,j}$ must be 0. Equation (25) and Equation (26) ensure that c_i and c_j do not overlap when they are in the same row. Equation (25) applies when c_i is to the left of c_j , while Equation (26) applies in the opposite situation. The boolean variable $g_{i,j}$ is used to distinguish between these two conditions. For instance, if $g_{i,j} = 0$, Equation (26) simplifies to:

$$x_i + w_i^c \leq x_j + M. \quad (27)$$

Since M is a constant much larger than all other numbers, Equation (25) becomes ineffective in restricting any variable. The term $M \cdot d_{i,j}$ in both equations ensures that Equations (25) and (26) only impose constraints when $d = 0$, meaning the cells are in the same row. Otherwise, the expressions will always hold.

By combining the constraint functions with the objective functions, the ILP formulation can theoretically find the optimal solution to the problem. This ensures that critical cells stay as close as possible to the TSPS, without significantly increasing the signal wire length, while meeting all design constraints.

C. Trapped Cell Relocation

The ILP formulation performs replacement such that cross-domain cells are relocated to be close to their TSPSs. To perform secondary power routing, the nearby cross-domain cells belonging to the same SPD are usually connected to each other, forming a local routing tree and then connecting to their common TSPS. This routing nature reduces wire length and conserves routing resources, but if a cell of

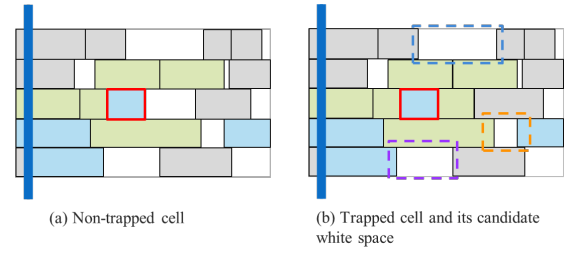


Fig. 6. Examples of trapped and non-trapped cells

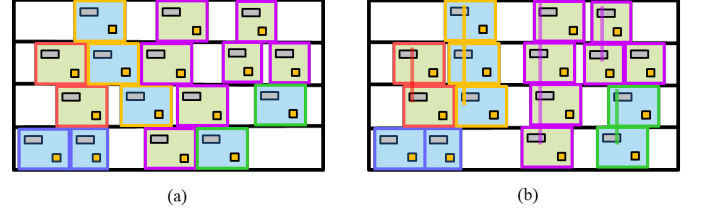


Fig. 7. Examples of pin alignment and clustering. (a) The cell clustering result before performing pin alignment, divided into four groups: blue, red, orange, and purple. (b) The cells after performing pin alignment, with line segments representing the planned routing paths

another domain is surrounded by these cells, it may be obstructed and suffer from routing difficulty, which will be labeled as a “trapped cell”. To avoid an excessively conservative mechanism, a cell can only be marked as trapped if there is no available space larger than its width in its vicinity. If there is a space larger than the cell width nearby. As shown in Fig. 6, the cell highlighted by red in Fig. 6(a) is not classified as trapped, and that in Fig. 6(b) is trapped.

Trapped cells often struggle with accessing secondary power nets, resulting in limited routing paths and potential connection violations. To address this, we propose an algorithm for relocating these cells. The algorithm employs a window-based approach to identify suitable spaces, starting with a square window equal to five rows in height. If no suitable space is found, the window expands until an appropriate location is identified. The found candidates are prioritized as follows: a space adjacent to a cell of the same SPD and is wider than the trapped cell (as the purple box in Fig. 6(b)) has the highest priority. The second choice is a space adjacent to a cell of the same SPD but narrower than the trapped cell (as the orange box), and the displacement of other cells to be shifted to accommodate the trapped cell should be smaller than a user-defined threshold. Finally, we consider a space that has no adjacent cells from the same SPD and is wider than the trapped cell (as the blue box).

D. Secondary Power Pin Alignment

The final stage involves a placement algorithm applied to all cross-domain cells collectively. After relocating the trapped cells, we aim to align the secondary power pins of all cross-domain cells belonging to the same SPD to a certain degree. This secondary power pin alignment step can contribute to routing bend and complexity minimization during secondary power routing.

The alignment process consists of two phases. The first phase involves cell clustering, as shown in Fig. 7(a), and the second phase focuses on aligning pins within each cluster, with the result shown in Fig. 7(b). Each cluster can contain up to ten cells. To group cells, we start with the bottom-left cell and look for other cells within the same SPD, prioritizing those closest to the secondary power pin. The cell closest to this pin is added to the cluster. After each addition, the centroid of the secondary power pins within the cluster is updated, and the search continues for the next nearest secondary power pin relative to the centroid. This process repeats until no suitable candidates remain or the cluster reaches its maximum size. It is crucial to note that this search has spatial restrictions. If there is any cell from a different SPD between a candidate cell and the cluster, or if the distance exceeds

TABLE II
EXPERIMENTAL RESULT OF WIRE LENGTH AND VIOLATIONS

Case2							Case3						
Situation	Dis	CriSecWL	SecWL	SignalWL	#Vios	CPU (s)	Situation	Dis	CriSecWL	SecWL	SignalWL	#Vios	CPU (s)
Original	10	310.66	15598	1324738	1340	0	Original	10	294.53	13817	1493950	1324	0
Only ILP		269.65	11604	1360978	591	2800	Only ILP		256.24	10059	1539067	605	2369
All techniques		270.31	11317	1373733	468	2803	All techniques		248.16	9974	1538907	498	2375
Original	20	406.86	16724	1399103	1338	0	Original	20	430.43	19764	1493825	1629	0
Only ILP		408.6	15409	1399640	671	2533	Only ILP		344.64	14035	1567163	675	2224
All techniques		409.69	15704	1405561	531	2536	All techniques		344.68	13827	1568370	524	2229
Original	30	599.81	28554	1324735	1937	0	Original	30	612.87	25627	1493803	1845	0
Only ILP		509.16	19564	1425908	806	1542	Only ILP		532.15	19684	1625694	763	1253
All techniques		515.23	20246	1441374	681	1547	All techniques		529.95	17487	1608271	632	1255
Original	40	789.94	40284	1326071	2105	0	Original	40	769.82	33112	1494328	2256	0
Only ILP		729.64	26487	1490347	768	1044	Only ILP		709.67	22654	1635195	703	1013
All techniques		732.96	26682	1461772	735	1047	All techniques		704.49	21867	1626657	698	1017
Average							Ratio						
Original		526.87	24185	1418819	1722	0	Original		1	1	1	1	-
Only ILP		469.96	17437	1505499	698	1847.25	Only ILP		0.89	0.72	1.06	0.4	1
All techniques		469.43	17138	1503080	596	1851.12	All techniques		0.89	0.7	1.05	0.34	1

one row vertically or two rows horizontally, the candidate will not be merged.

Once the clusters are determined, the secondary pin alignment algorithm is applied to each cluster, focusing on aligning the secondary power pins along the horizontal axis. The number of centroids used for alignment is based on the maximum number of cells in the same row within a cluster. For example, in the purple cluster in Fig. 7(b), the cells are divided into three groups, each aligning to a different centroid. The goal of this alignment is to ensure that the secondary power pins overlap near the center by at least half the width of a secondary power pin, making routing easier.

During the alignment process, if a cell movement is obstructed by adjacent cells, specific rules are followed. When only one cell blocks the movement, the adjustment is made but reduced to half the intended distance. However, if multiple cells block the movement, no adjustment is made for the cells in that row. After completing the alignment for all clusters, the centroids are recalculated, and the process is repeated. In this second alignment phase, any displacement of adjacent cells is again reduced by half. If the resulting displacement falls below a certain threshold, further movement is stopped.

IV. EXPERIMENTAL RESULTS

Our algorithm for the placement stage was implemented in C++ programming language. The routing stage is performed using the commercial tool INNOVUS [2] for routing and DRC verification. The complete framework was executed on a Linux machine equipped with a 3.5 GHz CPU and 72 GB RAM, running CentOS. We use IBM's CPLEX [3] as the ILP solver. To verify the effectiveness of the proposed methodologies, we utilize the benchmarks of the ISPD 2018 Contest on Initial Detailed Routing [1], which provides a legally placed design, including cell placement, circuit netlist, and technology constraints. Each design consists of 9 metal layers: the primary power stripes are placed on M9 and M8, and the secondary power stripes on M7 and M6. To meet advanced process requirements, layers M1 to M3 are set as 1D routing layers, while the remaining layers are available for 2D routing.

Regarding pin patterns, the original benchmark lacks secondary power pins, so we modify pin patterns to generate cross-domain cells. In our design, the cells within a window of 10×10 row heights are designated as cross-domain cells for secondary power routing, comprising approximately 2% to 12% of all cells. Each design includes two secondary PDNs. The secondary power routing follows a trunk-based pattern with a fanout limit of 10. After routing the benchmark circuits (case1 through case10) in the routing stage, we observed that case2 and case3 experienced significant numbers of additional violations due to secondary power routing. Consequently, these two cases are used as primary targets for presenting experimental results, as they feature higher pin densities and a greater proportion of cross-domain cells compared to other designs.

Empirically, the distances between cross-domain cells and their TSPSs can significantly impact overall design quality. In designs with larger secondary power stripe spacings, moving critical secondary cells closer to their target stripe becomes a priority. Conversely, in designs with smaller stripe spacings, the focus shifts to fine-tuning pin alignment. To demonstrate the effectiveness of our algorithm across various secondary power stripe spacings, we conduct experiments with different stripe distances as a variable. In addition, we limit each ILP iteration to 100 seconds, and the parameter α in Equation (2) is set to 0.99, prioritizing aggressive reduction of secondary PDN wire length over preservation of signal wirelength.

The experimental results are shown in Table II. In the table, "Dis" refers to the spacing between secondary power stripes, measured by the number of rows in between. "CriSecWL" is the critical secondary PDN wire length, calculated by summing the longest 10 secondary power nets to provide stable data without large fluctuations. "SecWL" is the total wire length of the secondary PDNs, and "SignalWL" is the total wire length of signal nets. "#Vios" counts the number of design rule violations, and "CPU" represents the total runtime during the placement stage. In addition, we compare three scenarios: without using the placement algorithm ("Original"), using only the ILP formulation for placement refinement ("Only ILP"), and using the complete placement algorithm ("All techniques"). This comparison tests the effectiveness of the trapped cell relocation and pin alignment components through an ablation study.

The experimental results show that the ILP-based placement refinement approach averagely reduces the critical secondary PDN wire length by 11% and the total secondary PDN wire length by 28%, with only a 6% increase in signal wire length. Most importantly, the replacement significantly reduces the violations by 60% on average. Furthermore, incorporating trapped cell relocation and pin alignment into the optimization flow can further achieve 2% and 1% reductions in the PDN wire length and the signal wire length, and a 6% reduction in the number of violations, with minimal impact on both critical secondary wire length and signal wire length, further enhancing the overall algorithm performance.

V. CONCLUSIONS

This paper presents a secondary-power-routing-aware detailed placement algorithm that facilitates the repositioning of cross-domain cells, aiding multiple power domain designs in executing secondary power routing smoothly. The algorithm significantly reduces the secondary PDN wire length and mitigates open circuit issues caused by routing failures. As a result, the total secondary PDN wire length decreased by 30%, the critical secondary PDN wire length was reduced by 11%, and the average number of violations decreased by 66%.

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